EXPERIMENT NO. 6

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Input Code:

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* top = NULL;
void insertNode() {
    int val;
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        return;
    }
    printf("Enter value: ");
    scanf("%d", &val);
    newNode->data = val;
    newNode->next = top;
    top = newNode;
    printf("Inserted %d\n", val);
}
void deleteNode() {
    if (top == NULL) {
        printf("Stack Underflow\n");
        return;
    }
```

```
    struct Node* temp = top;
    printf("Deleted: %d\n", temp->data);
    top = top->nex        t;
    free(temp);
}
void display() {
    if (top == NULL) {
        printf("Stack is empty\n");
        return;
    }
    struct Node* ptr = top;
    printf("Stack elements: ");
    while (ptr != NULL) {
        printf("%d ", ptr->data);
        ptr = ptr->next;
    }
    printf("\nTop element: %d\n", top->data);
}
int main() {
    int choice;
    do {
        printf("\n1. Insert\n2. Delete\n3. Display\n4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                insertNode();
                break;
            case 2:
```

```
            deleteNode();
            break;
        case 3:
            display();
            break;
        case 4:
            break;
        default:
            printf("Invalid choice\n");
    }
  } while (choice != 4);
  return 0;
}
```

Output:

```
1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 1
Enter value: 11
Inserted 11

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 1
Enter value: 12
Inserted 12

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 1
Enter value: 13
Inserted 13

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 3
Stack elements: 13 12 11
Top element: 13
```

```
1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 2
Deleted: 13

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 3
Stack elements: 12 11
Top element: 12

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 4
```